MSC CYBERSECURITY & DATA SCIENCE · ESAIP ANGERS · GROUP 2

TrustNav

Can a fleet believe its satellite position?

A trust score from 0 to 100 for every vehicle, every second. Explained in plain words. Acted on before a lie reaches dispatch.



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Tutor: Sofiane Hamrioui · Oral examination, 13 November 2026

THE PROBLEM

Everything that moves trusts GNSS blindly

1 500

flights spoofed per day around the Baltic and conflict zones, summer 2024. A daily event for aviation since 2022.



Nantes airport, 2017
A jammer left running in a parked car delayed several flights. Fine: €2 000.



Lyon airport, 2023
A truck driver's jammer, used to escape fleet tracking, disrupted signals until the ANFR traced it.

WHAT EXISTS TODAY

Fleet telematics
shows whatever position it is given

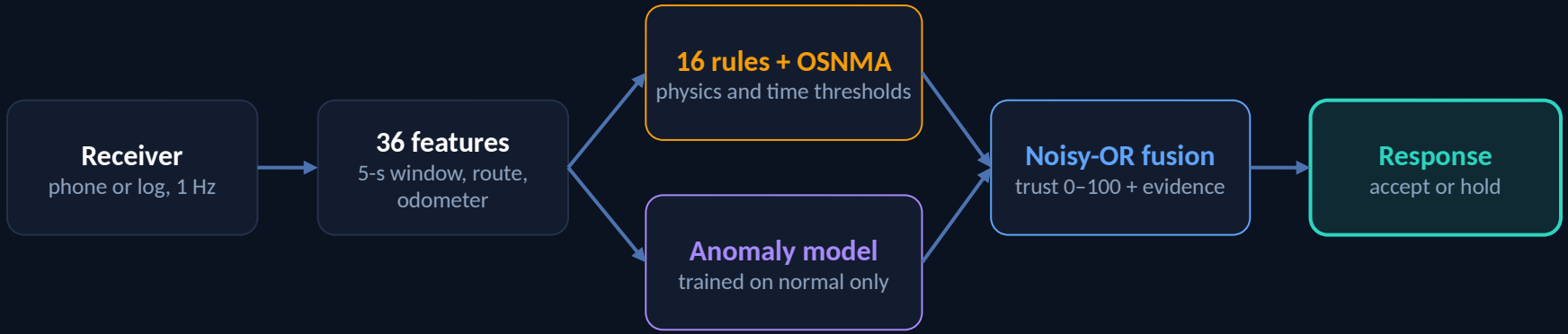
Receiver flags
a state code, no explanation

Fixed probes, anti-jam antennas
protect a site, not a moving van

The gap Nobody tells the operator, per vehicle and per second, whether the position is true, or acts on it.

THE IDEA

A software layer beside the receiver



Scores

0 to 100, per vehicle and drone, every second

Explains

in plain words, with values: “Fix moved 7 998 m in 1 s (limit 250 m)”

Responds

refuses the fix and tracks the vehicle by its odometer along its route

DETECTION

Three witnesses, one verdict

Rules

16 checks with named thresholds: motion, signal and time

CATCHES a van cannot jump 250 m in a second

MISSES slow attacks never cross a threshold

60 m/s · 8 m/s² · 6 dB-Hz C/N0 shift · time ±5 s

Anomaly model

Isolation Forest + tail detector, trained on normal driving only

CATCHES a fix 90 m off the road with clean signals

MISSES noisier; must be calibrated on the receiver

risk = 25 · (−log10 p): 50 is a 1-in-100 event

Galileo OSNMA

satellites sign their navigation data (operational since July 2025)

CATCHES forged data: the signature fails

MISSES relay attacks keep it valid; GPS not covered

one witness among three, not a defence

Fused by noisy-OR, not averaged

risk = 100 · [1 − (1 − 0.6 · p rules)(1 − 0.8 · p model)] trust = 100 − risk

MODEL SAYS 90
RULES SAY 20

55 average dilutes it

75 noisy-OR keeps it

The response layer: the lie stops here

1

ACCEPT

trust at least 50 and no critical rule: the fix passes through unchanged.

2

HOLD

the fix is refused. The position handed on is dead-reckoned from the last trusted fix with the odometer, along the planned route. A spoofer can touch neither.

3

RELEASE

after 5 clean seconds and a fix within 60 m + 3 % of the distance travelled. After 60 s the bound is ten times looser; a fix still kilometres away stays refused.

CONTINUE VERIFY ISOLATE



The console during the teleport: the van's real path (left), the reported position 8 km away (right), and the dashed lie between them.



What the operator reads

Score 15, Untrusted. Rules 45, model 100, combined risk 85.

"The receiver places the van 8.0 km from the road it is actually on… TrustNav has stopped passing this position on."

EVALUATION

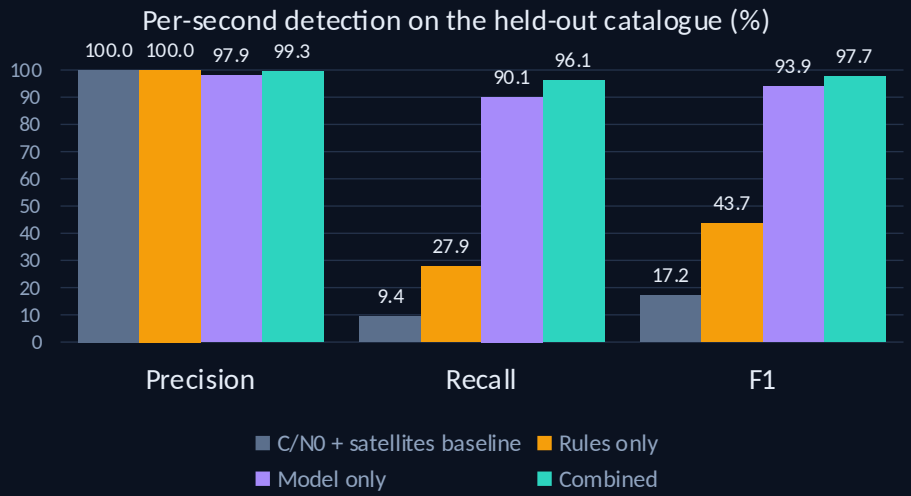
Tested on routes it has never seen

99.3 %
PRECISION

96.1 %
RECALL

0.44 %
FALSE ALARMS

1 s
MEDIAN DELAY



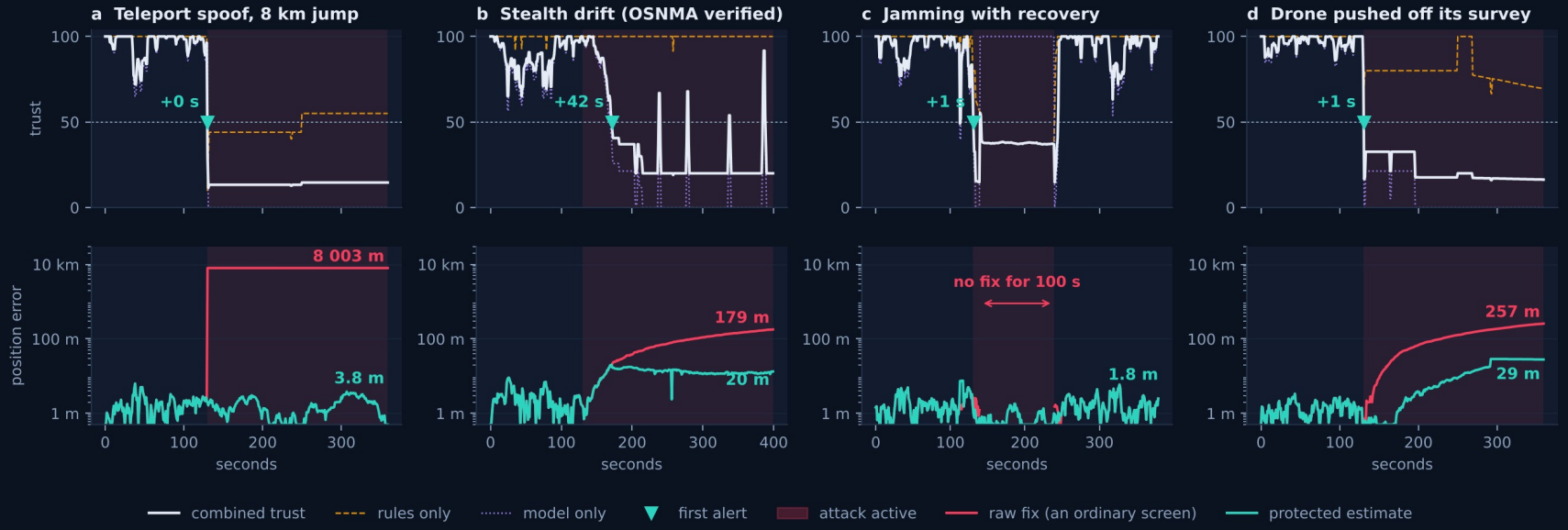
10 scenarios · 3 800 seconds
1 550 under attack, 7 attack types, real Angers streets

One role per route
train, validate or test: never shared

95 % intervals over scenarios
precision 97.9–100 · recall 89.4–99.6

FOUR HELD-OUT ATTACKS

What happens, second by second



7 / 7
attack episodes detected

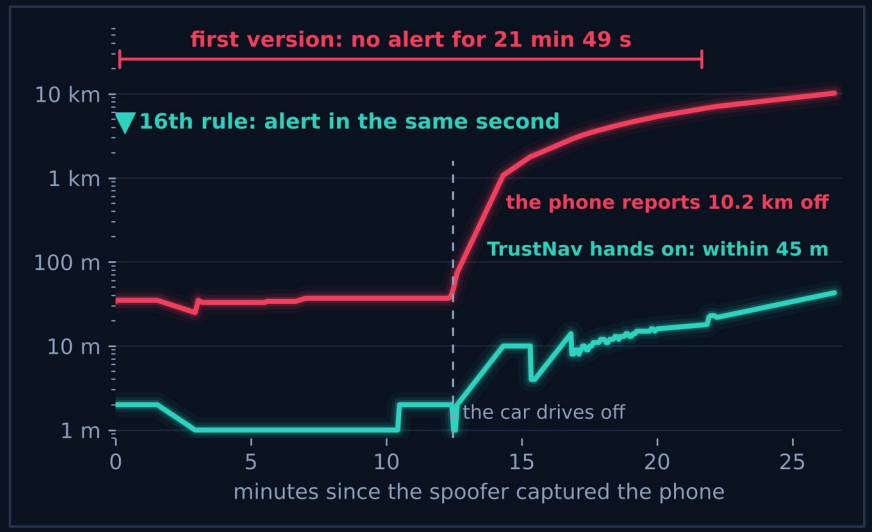
8 km → 29 m
worst position handed on

100 s → 0 s
blank screen under jamming

42 s
stealth drift caught, 12 m off the road

BEYOND THE SIMULATOR

Then we replayed real attacks



Jammertest 2024, Andøya, Norway: the authorities let jammers and spoofers transmit. A phone in a car is captured for 26.5 min; the u-blox beside it is not, and gives the truth. Recordings: VAARA dataset, University of Vaasa. Handed-on error is a best case (reference speed as odometer).

21 min 49 s
before the version we had published raised an alert
It refused 5 of 909 captured fixes: nothing in the spoofed stream was physically impossible.

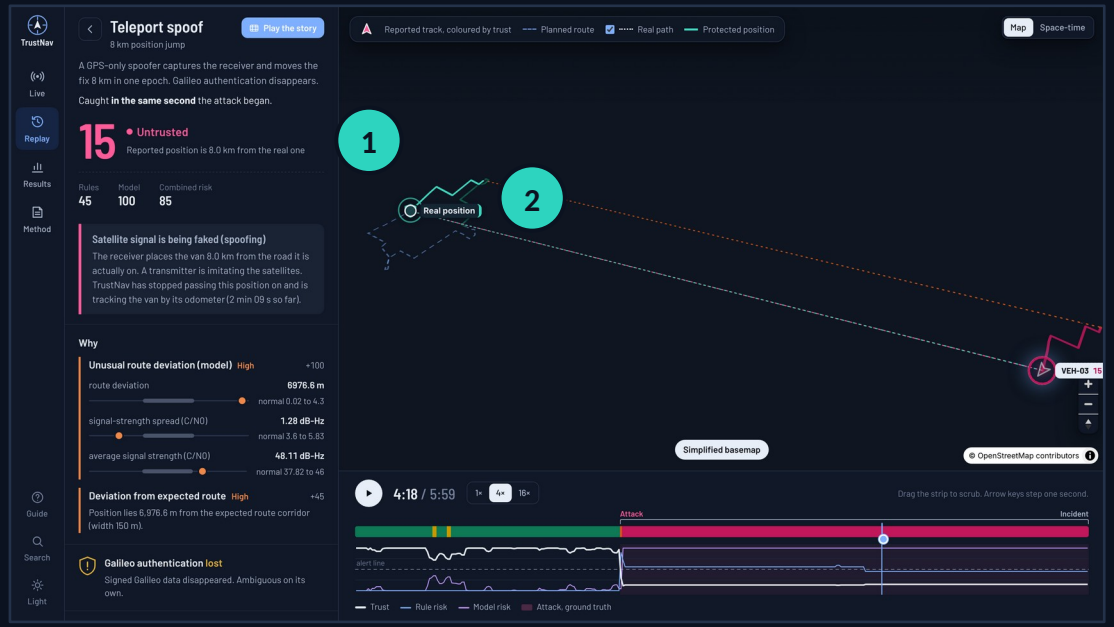
same second
with a 16th rule: receiver time against the device's own clock
909 of 909 captured fixes refused. No false alert in 37 clean minutes.

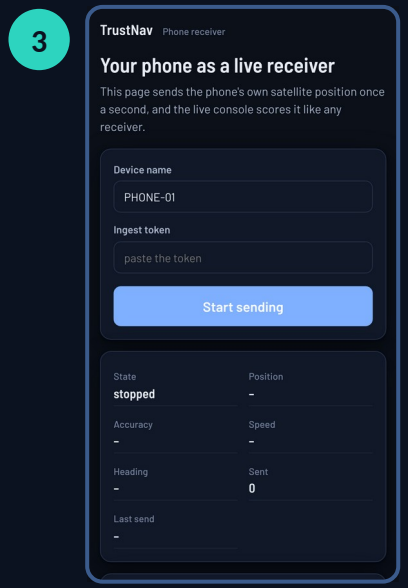
15 of 20
real jamming exposures alerted
A u-blox driving past real jammers. Rules alone: 1 of 20, its NMEA carries no jamming indicator.

The first version failed. We found why, fixed it, measured it.
A clean real drive, 27.6 min: no false alarm from the rules

LIVE DEMO

The console, and a phone as a receiver





1 score, verdict, evidence **2** real path vs the reported lie **3** your phone, scored live, with an attack switch

trustnav.vercel.app · open source · Python engine, 42 tests · Supabase, row-level security · Next.js console, 109 end-to-end checks

HONEST CLAIMS

What we claim, what we don't, what's next

Limits

- simulator numbers are an upper bound
- real attacks: two days of one campaign
- the response is as good as its odometer
- OSNMA simulated; no multi-antenna evidence

Safety and ethics

- nothing transmitted: software and recordings
- future hardware is receive-only
- GDPR: purpose, 3-day retention, transparency
- a research indicator, not certified integrity

Next

- hours of real drives in Angers
- raw I/Q captures: Jammertest 2023, TEXBAT
- a second receiver against synchronous spoofing
- real OSNMA from raw Galileo pages

A per-second, explained trust score, and a response that keeps a lie from the systems that act on it.

trustnav.vercel.app · github.com/domail2karandeshmukh-cyber/trustnav

Questions?